

QUESTIONS 8 / 8 completed

Character ▾

CHARACTER

You are developing a game that involves different types of characters, such as heroes, villains, and monsters. Write a Java program to create an interface called "Character" with methods for moving, attacking, and taking damage. Implement the interface in classes for each type of character, such as "Hero", "Villain", and "Monster".

PROGRAM EDITOR

Java ▾

Run

Save

```

1 interface Character {
2     void move();
3     void attack();
4     void takeDamage();
5 }
6
7 class Hero implements Character {
8     public void move() {
9         System.out.println("Hero moves.");
10    }
11
12    public void attack() {
13        System.out.println("Hero attacks.");
14    }
15
16    public void takeDamage() {
17        System.out.println("Hero takes damage.");
18    }
19 }
20
21 class Villain implements Character {

```

INPUT

Your INPUT goes here!

OUTPUT

```

Compilation Error:
C:\Windows\TEMP\sandbox_192572089RC_6a6ae416bba8a0.36087736\Hero.java:49: error:
class GameCharacters is public, should be declared in a file named
GameCharacters.java
public class GameCharacters {
    ^
1 error

```

QUESTIONS 8 / 8 completed

Sort

SORT

Write a Java program to create a list of integers and sort them in ascending order using the Collections framework.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.ArrayList;
2 import java.util.Collections;
3 public class SortList {
4     public static void main(String[] args) {
5         ArrayList<Integer> numbers = new ArrayList<>();
6         numbers.add(45);
7         numbers.add(12);
8         numbers.add(78);
9         numbers.add(23);
10        numbers.add(5);
11        System.out.println("Original List: " + numbers);
12        Collections.sort(numbers);
13        System.out.println("Sorted List (Ascending): " + numbers);
14    }
15 }
```

OUTPUT

```
Original List: [45, 12, 78, 23, 5]
Sorted List (Ascending): [5, 12, 23, 45, 78]
```

INPUT

Your INPUT goes here!

QUESTIONS 8 / 8 completed

HashSet

HASHSET

Write a Java program to create a HashSet of strings and remove a specific element from it.

PROGRAM EDITOR

Java



Run

Save

```
1 import java.util.*;
2 public class RemoveElement {
3     public static void main(String[] args) {
4         Set<String> names = new HashSet<String>();
5         names.add("Apple");
6         names.add("Mango");
7         names.add("Orange");
8         names.add("Banana");
9         System.out.println(names);
10        names.remove("Orange");
11        System.out.println(names);
12    }
13 }
```

OUTPUT

Security Error: Disallowed code pattern detected.

INPUT

Your INPUT goes here!

QUESTIONS 8 / 8 completed

TreeMap

TREEMAP

Write a Java program to create a TreeMap of strings and integers and iterate over its entries.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.TreeMap;
2 import java.util.Map;
3 public class TreeMapExample {
4     public static void main(String[] args) {
5         TreeMap<String, Integer> students = new TreeMap<>();
6         students.put("Alice", 85);
7         students.put("Bob", 90);
8         students.put("Charlie", 78);
9         students.put("David", 95);
10        System.out.println("TreeMap entries:");
11        for (Map.Entry<String, Integer> entry : students.entrySet()) {
12            System.out.println(entry.getKey() + " = " + entry.getValue());
13        }
14    }
15 }
```

OUTPUT

```
TreeMap entries:
Alice = 85
Bob = 90
Charlie = 78
David = 95
```

INPUT

Your INPUT goes
here!

QUESTIONS 8 / 8 completed

PriorityQueue ▾

PRIORITYQUEUE

Write a Java program to create a PriorityQueue of integers and add elements to it.

PROGRAM EDITOR

Java ▾

Run

Save

```
1 import java.util.PriorityQueue;
2 public class PriorityQueueExample {
3     public static void main(String[] args) {
4         PriorityQueue<Integer> numbers = new PriorityQueue<>();
5         numbers.add(30);
6         numbers.add(10);
7         numbers.add(50);
8         numbers.add(20);
9         numbers.add(40);
10        System.out.println("PriorityQueue: " + numbers);
11    }
12 }
```

OUTPUT

```
PriorityQueue: [10, 20, 50, 30, 40]
```

INPUT

Your INPUT goes
here!

QUESTIONS 8 / 8 completed

LinkedListMap ▾

LINKEDHASHMAP

Write a Java program to create a LinkedHashMap of strings and integers and retrieve the keys in the order they were inserted.

PROGRAM EDITOR

Java ▾

Run

Save

```
1 import java.util.LinkedHashMap;
2 public class LinkedHashMapExample {
3     public static void main(String[] args) {
4         LinkedHashMap<String, Integer> students = new LinkedHashMap<>();
5         students.put("Alice", 85);
6         students.put("Bob", 90);
7         students.put("Charlie", 78);
8         students.put("David", 95);
9         System.out.println("Keys in insertion order:");
10        for (String key : students.keySet()) {
11            System.out.println(key);
12        }
13    }
14 }
```

OUTPUT

```
Keys in insertion order:
Alice
Bob
Charlie
David
```

INPUT

Your INPUT goes
here!

QUESTIONS 8 / 8 completed

LinkedHashMap ▾

LINKEDHASHMAP

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PROGRAM EDITOR

Java ▾

Run

Save

```
1 import java.util.LinkedHashMap;
2 public class LinkedHashMapExample {
3     public static void main(String[] args) {
4         LinkedHashMap<String, Integer> students = new LinkedHashMap<>();
5         students.put("Alice", 85);
6         students.put("Bob", 90);
7         students.put("Charlie", 78);
8         students.put("David", 95);
9         System.out.println("Keys in insertion order:");
10        for (String key : students.keySet()) {
11            System.out.println(key);
12        }
13    }
14 }
```

OUTPUT

```
Keys in insertion order:
Alice
Bob
Charlie
David
```

INPUT

Your INPUT goes here!

QUESTIONS 8 / 8 completed

Max-Min

MAX-MIN

Write a Java program to create a Set of integers and perform different operations on it, such as finding the maximum and minimum values.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.HashSet;
2 import java.util.Set;
3 import java.util.Collections;
4
5 public class SetOperations {
6     public static void main(String[] args) {
7         Set<Integer> numbers = new HashSet<>();
8
9         numbers.add(45);
10        numbers.add(12);
11        numbers.add(78);
12        numbers.add(23);
13        numbers.add(5);
14
15        System.out.println("Set: " + numbers);
16        System.out.println("Maximum Value: " + Collections.max(numbers));
17        System.out.println("Minimum Value: " + Collections.min(numbers));
18    }
19 }
20
```

OUTPUT

```
Set: [5, 23, 12, 45, 78]
Maximum Value: 78
Minimum Value: 5
```

INPUT

Your INPUT goes
here!